

Otago Challenge 2012 Set 1

1 Problems

Problem 1.1 Jack multiplies a number by 2 and gets 60 as his answer. However, he should have divided the number by 2 to get the correct answer.

What is the correct answer?

Problem 1.2 When a Sudoku grid is completed, every column, every row and each of the four 2×2 boxes (outlined) contains the digits 1 to 4.

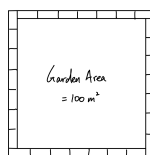
Complete all the remaining entries in the Sudoku grid shown.

1	2	3	
	4	2	1

Problem 1.3 We can call this year, 2012, a *sum-product* year, because the sum of the first two digits is equal to the product of the last two digits. 2021 is also a *sum-product* year. What is the next year after 2021 which is again a *sum-product* year?

Problem 1.4 Twenty eight rectangular paving stones, all the same size, form a path around a square garden of area 100 square metres, as shown (but not to scale).

Each paving stone is three times as long as it is wide. What is the length of a paving stone?



Problem 1.5 Emma's dad took part in a triathlon, in which each contestant swam 2km, cycled 30km and ran 8km. During the triathlon he swam at 2.5km per hour, cycled at 25km per hour and ran at 12km per hour.

What was his average speed (in km per hour) over the whole triathlon course?